# THE CONSTRUCTOR
# MATHEMATICAL JOURNAL

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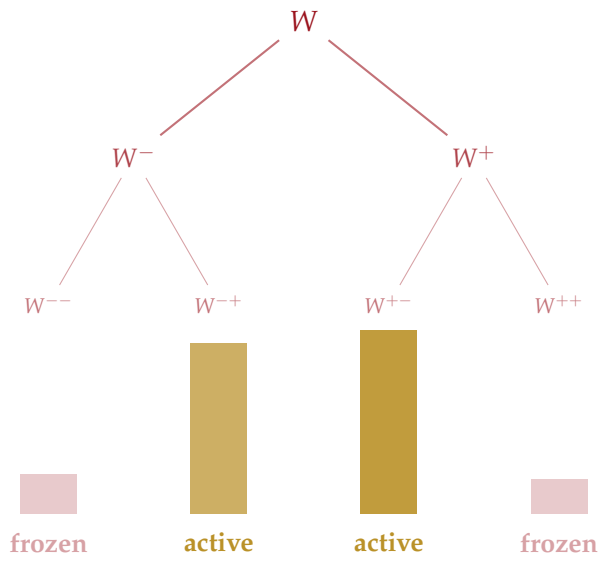


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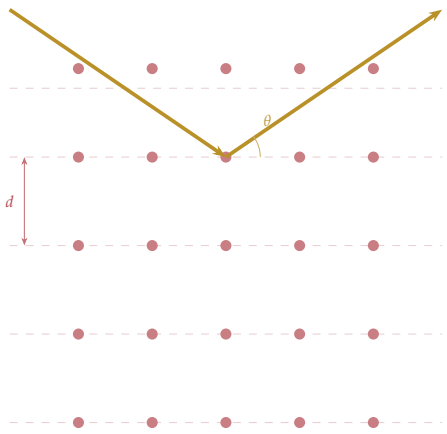


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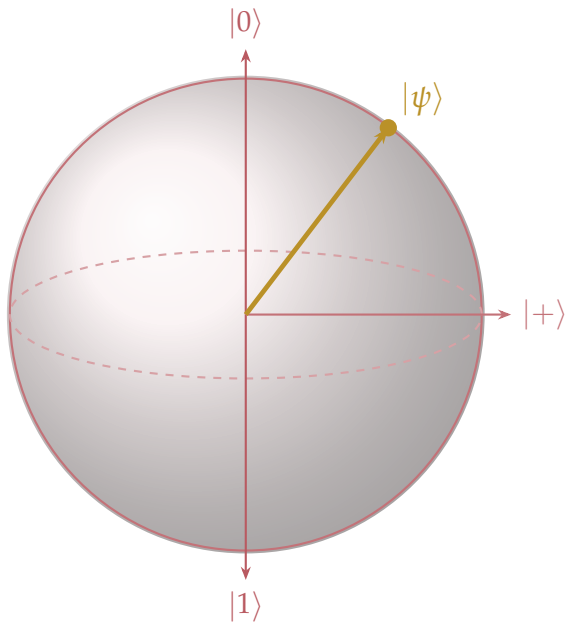


*Bloch Sphere*

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## The Journal

The *Constructor Mathematical Journal* (**CMJ**) is the undergraduate mathematics journal of Constructor University Bremen, published bi-annually by the Mathematics Society at Constructor University (MSCU).

The CMJ publishes expository and research articles written by undergraduate students. All articles originate in the MSCU's own Mathematics Undergraduate Seminar (UgS) and are developed into self-contained expositions, accessible to a broad mathematical audience.

## Recorded Talks

Each article in the CMJ is accompanied by a recorded talk delivered at the UgS. The recordings are freely available and provide an accessible entry point to each paper's ideas.

`elshenawyom.github.io/teaching/ugs`

## Editorial Board

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## Cover Art

Bragg's Law on a protein crystal lattice (left) reflects the rays that determine its structure; a Bloch sphere represents the qubit (centre) that is a superposition of ones and zeroes; and a channel polarization tree (right) freezes channels into reliable and unreliable ones to realise Shannon's capacity.

## Submission Guidelines

We welcome submissions from students at Constructor University and from visiting students who have participated in the UgS. Manuscripts should:

- be written in LaTeX using the `cmj` style package (available at the journal website);
- include an abstract of 80–150 words;
- be self-contained and aimed at an undergraduate audience;
- include a complete reference list.

To participate in the Mathematics Undergraduate Seminar (UgS), there is a simple application process. One simply attends to a member of the MSCU board, possible through the respective contact channels, or closely follows the MSCU's latest announcements.

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# On the CMJ's First Volume

## A Short Story

The Mathematics Undergraduate Seminar (UgS) is a tradition of student research and exposition, first introduced as part of the MATH programme. This tradition has persisted for years on, until the Spring of 2024. The programme was restructured into an MMDA — Mathematics, Modeling & Data Analytics — variant, and the seminar was discontinued.

As a seminarist myself, this tradition is encoded in Constructor (well, Jacobs...) University's mathematical DNA. A discontinuation in itself would be a step backwards for its mathematics society. So, we relaxed the curricular condition and took the seminar outside the classroom. The UgS was therefore reborn as a student-led initiative, attracting a superset of mathematics and non-mathematics students.

`elshenawyom.github.io/teaching/ugs`

In its incarnation, the seminar is now a thriving hub of mathematical exploration, bringing students and faculty closer than ever before. This journal is the fruit of this rebirth.

## An Honourable Mention

It is our honour and pleasure to acknowledge the following (MATH'27) students,

- **Klesta Qehaja**, for her work on *Random Walks in* $\mathbb{Z}^d$,
- **Pablo Santos Guerrero**, for his work on *Piccard-Vessiot Theory and Liouville's Theorem*,
- **Vano Kakiashvili**, for his work on *the Legendary 1988 IMO Problem Six* .

Despite not accumulating into full articles, their contributions to the UgS are nothing short of excellent. The inauguration of this journal would not have been possible without you.

## Acknowledgements

We would like to express our sincere gratitude to the Mathematics faculty, for the warm support and encouragement. In particular, we thank Prof. Dr. Keivan Mallahi-Karai, for peerlessly advising this volume and for his unwavering support of the UgS. Worthy of mention is Dr. Ivan Ovsyannikov, for both his academic and moral contributions to the seminarists. It means a lot for us students.

# Editor's Address

Dear reader,

It is with great pleasure that we present the inaugural edition of the *Constructor Mathematical Journal*, the student journal of the Constructor University Mathematics Society.

This journal was born from a desire to give undergraduate researchers a venue to communicate their work and to foster a culture of mathematical inquiry at Constructor University Bremen. Each article collected here originated in our Mathematics Undergraduate Seminar (UgS) and has been developed into an expository piece intended to be accessible to a wide audience.

In its very-first volume, the CMJ proudly presents to you three excellent articles,

- **Shannon's Theorem and Polar Codes** explores the foundations of reliable communication laid by Claude Shannon and traces the path to Erdal Arıkan's breakthrough construction of polar codes.
- **Bragg's Law in the Context of Protein Structure Determination** traces back to W. L. Bragg's 1913 discovery, on to the modern protein structure determination pipeline, illustrating how the prior enables the recovery of atomic-scale structure from diffraction data.
- **The Qubit and Shor's Algorithm** provides a step-by-step journey from the basics of quantum computing to Peter Shor's celebrated factoring algorithm and its implications for modern cryptography.

As I sign-off as Editor-in-Chief, I am excited to see how this journal evolves in the coming years. To the seminarists, I invite you to pass on the torch to the next generation of students, to explore fascinating mathematics through the UgS, to share that same fascination through the CMJ. To the students, I encourage you to get involved in this exciting venture that is the UgS. There you will always find a place to be curious, to ask questions, to make mistakes and learn from them, to grow as a mathematician, and most importantly, as a person.

The CMJ is a testament to the mathematical excellence of those who contributed to it, and to the potential of those courageous enough to try. May it transcend the brevity of our time and echo softly in the spaces we leave behind.

*Yours Sincerely,*
**Omar Elshinawy — MATH'25**
Editor-in-Chief of the CMJ

# Shannon's Theorem and Polar Codes

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**Abstract.** Coding Theory is responsible for facilitating reliable message-passing over communication channels. In 1948, Claude Shannon proved that for every channel there exists a family of error-correcting codes that allow reliable communication at any rate below the channel's capacity. Shannon's work is, however, non-constructive, as it provides no method to design such efficient codes. While there have been attempts over the years to reach this theoretical capacity, notably with LDPC codes (1990) and Turbo Codes (1993), these could only empirically approach the theoretical limit. This stood for 60 years until Erdal Arıkan's introduction of Polar Codes, the first clean and mathematically proven family of codes to achieve Shannon's maximum capacity over symmetric binary-input memoryless channels. This expository paper aims to briefly introduce the general Coding Theory scheme, outline the key ideas in Shannon's arguments, and explore Arıkan's outstanding construction of polar codes.

## 1 Coding Theory Fundamentals

Before proceeding we need to build a framework of definitions and the general idea behind Coding Theory. This paper is designed such that anyone with a basic background in mathematics can understand. The details of what information and entropy are have been moved under the section 'Shannon's Theorem' for convenience as they follow directly into his idea of relating channel capacity with information rate of a code.

In real life, messages are sent as a block of symbols through some medium, aka a 'channel'. For example, binary code.

**Channel:** a formal mathematical model of communication that operates abstractly like a function, in the sense that it has input and output sets. Channels apply a probabilistic function on each symbol of the input to produce the output. The set of all symbols is the alphabet $\Sigma$

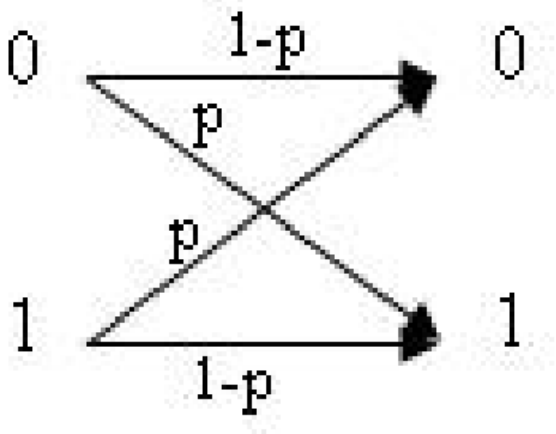



Figure 1: Model of a $BSC_p$

In our case we use the Binary Symmetric Channel.

**Binary Symmetric Channel (BSC):** both input and output sets consist of the alphabet $\Sigma = \{1, 0\}$. The probabalistic function is the Bernoulli random variable with probability $p$ of being flipped as visualised in Figure 1 and Equation 1. The Bernoulli random variable is hereafter denoted as as $X \sim Bern(p)$.

$$u = \begin{cases} u & 1-p \\ u' & p \end{cases} \tag{1}$$

A more generalized channel is also the discrete memoryless channel. However, we will not be discussing it in this paper. **Discrete Memoryless Channel:** Channels that transmit sets of symbols from a finite alphabet $\Sigma$. *Memoryless* simply means that the probabilistic function is independent of messages sent before the current message.

Now that errors have been modeled, we must find a way to correct or detect them. Here we introduce the notion of codes through the below definitions.

**error-correcting code:** an algorithm that represents messages of length $k$ as codewords of length $n$ where $n > k$ such that the occurance of errors after transmission can be detected or corrected.

**message:** a sequence of symbols $\mathbf{u} = u_1 u_2 ... u_k$. It is the information we desire to transmit.

**codeword:** A sequence of symbols $\mathbf{x} = x_1 x_2 ... x_n$ where each codeword corresponds directly to a message $\mathbf{u}$ as $\mathbf{x} = \mathbf{x}(\mathbf{u})$.

The set of all messages and codewords are $U$ and $\chi$, where $|\chi| = |U| = 2^k$

More specifically, we are looking at Linear Codes, the parent family of Polar Codes.

| $x_1$ | $x_2$ | $x_1$ XOR $x_2$ |
|---|---|---|
| **0** | 0 | 0 |
| **0** | 1 | 1 |
| **1** | 0 | 1 |
| **1** | 1 | 0 |

Figure 2: XOR truth table

## 1.1 Definitions and Properties of linear codes

**Definition 1:** A code is linear if there exists a matrix $H$ such that $Hx^T = 0 \quad \forall u$. This is called the parity check matrix.

(a) Additionally, $H(x+y)^T = Hx^T + Hy^T = 0$ and so $x+y$ is also considered a codeword. This is the linearity property of linear codes.

**Definition 2:** For every linear code there exists a $k$ x $n$ generator matrix $G$ such that $\mathbf{x} = \mathbf{u}G$ and $GH^T = HG^T = 0$.

**Definition 3:** The information rate (or efficiency) of a code is $R = \frac{k}{n}$.

The matrices and properties above can be used directly to encode and decode simple codes via the H and G matrix pair.

**Definition 4:** The Hamming weight is the count of '1's present in a base 2 vector. It is denoted by $wt(\mathbf{x})$

**Definition 5:** The Hamming distance between any 2 vectors $\mathbf{x}$ and $\mathbf{y}$ is defined as the number of differences between them. The notation is as follows: $dist(\mathbf{x}, \mathbf{y}) = wt(\mathbf{x} - \mathbf{y})$

**Definition 6:** The minimum distance of a code is the minimum Hamming distance between any two codewords. It is often parameterized as $d$. This is simply referred to as the *distance* of a code.

**Definition 7:** the XOR operation between any two bits is denoted X + Y where X and Y are bits, and is defined in Figure 2

Finally, we can describe simple binary codes as a set of parameters $[n, k, d]$, encapsulating the

message and codewords size, alongside the minimum distance between each codeword.

To better understand the minimum hamming distance of a code, recall that the size of the set of codewords and messages are equal, yet the length of each codeword is longer than the original message. This means the set of codewords $\chi \subset \{0,1\}^k$, leaving gaps in between.

We can visualize this using Hamming balls in Figure 3. The center represents a valid codeword and the radius outlines the area containing other elements of $\{0,1\}^k$ that aren't codewords. These are the elements obtained due to some error during transmission.

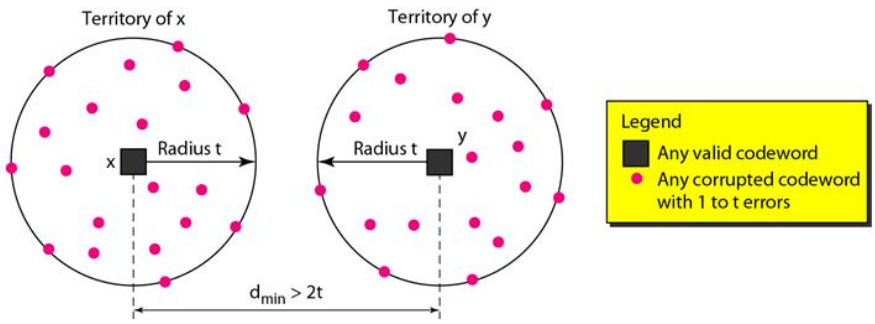



Figure 3: Visualization of Hamming balls with radius t and minimum distance of 2t. x and y are both codewords

Encoding is a simple process of just applying the generator matrix. We now explore one simple decoding method that will also provide insight into what the minimum distance truly means and its importance in maintaining a stable code.

## 1.2 Minimum distance decoding

Given a codeword $x$ sent over a BSC such that the received codeword is $y$ and probability of error is $0 \leq p < \frac{1}{2}$, the error vector is $e = x - y$. It describes the differences between sent and received codewords. The number of errors, aka the weight of e, is denoted $a$. In an ideal case, for example, $e$ would be a zeros vector with $a = 0$ .

We can find the probability of each error as $Pr(e = w) = p^a(1-p)^{n-a}$. Where $w$ is some error vector.

By observing the pattern as $a$ increases, we see that the most probable error vector (that is non-zero) is one with the least number of errors. As $(1-p)^n > p(1-p)^{n-1} > p^2(1-p)^{n-2} > \ldots$ due to our bounds on $p$, we can see that the most probable error vector (that is non-zero) is one with the least number of errors. Using this, we can test all $2^k$ error vectors in order from smallest to greatest error until we find a value for $x$ that is within the set of all valid codewords $\chi$.

In minimum distance decoding, the radius of the hamming ball is the maximum number of errors that can occur and still be corrected. If there is an overlap between hamming balls, then there is the possibility of an error occurring where it could be corrected to either valid codeword. By introducing redundancy, we increase the minimum distance to reduce this overlap, but as a result reduce the efficiency of the code.

For example, take codewords $c_1$ and $c_2$. if the distance between them is 1, we can only detect an error of distance 1 using the parity matrix, but cannot correct it with certainty. This is because we cannot determine which codeword the error originated from. However, if we have a distance of 2, we can detect errors up to distance 2 and we can correct error up to a distance of 1. This can be visualized as hamming balls with a distance between codewords of 2, and radius of 1, resulting in no overlap and is thus stable.

## 1.3 Extra Mathematics

The following mathematical ideas will help in our understanding of Shannon's Theorem but are not essential for the paper.

### 1.3.1 Law of Large Numbers (LLN)

The LLN states that as the sample size (n) of a random variable increases, the sample mean converges to the expected value (population mean).

Let $X_1, X_2, \ldots, X_n$ be independent and identically distributed (i.i.d.) random variables with finite mean $\mu = \mathbb{E}[X_i]$. The average is

$$\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

Weak Law of Large Numbers: For any $\epsilon > 0$,

$$\Pr\left(\left|\overline{X}_n - \mu\right| > \epsilon\right) \to 0 \quad \text{as } n \to \infty.$$

This means that the sample average converges to $\mu$ in probability.

### 1.3.2 Chernoff Bounds

Chernoff bounds help us find the probability of obtaining a large value in a random variable.

Imagine you are flipping a biased coin $n$ times. You expect to see $np$ heads, but in reality it could be within some range of $np$.

Chernoff bounds state that the probability of obtaining a value far from $np$ decreases exponentially as $n$ increases.

This is different from the Law of Large numbers, which would instead state that the fraction of heads to tails would approach $p$.

This probability is formally defined as:

$$P(X \geq a) \leq e^{-at} M_x(t) \qquad \forall t > 0$$

Where $M_x(t)$ is the Moment Generating Function and $a = (1+\delta)\mu$ as seen in 4.

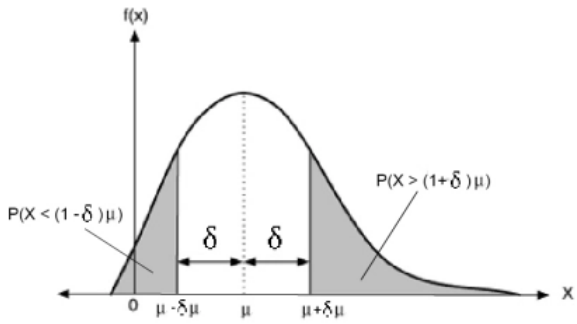



Figure 4: Probability distribution of a random variable

## 2 Examples of Linear Codes

## 3 The Coding Problem

The coding problem is the challenge of designing codes that can efficiently and reliably transmit data over noisy channels. To do this, we try to maximize the rate of transfer $R$ which involves increasing $k$ and reducing $n$ while simultaneously increasing the distance $d$ to allow us to correct more errors.

### 3.1 Example

Let's give a simple example of us encoding and decoding by talking. We define a Discrete Memoryless Channel (DMC) where the alphabet is the entire dictionary, so every word is a symbol. Then we define our noise function as the following: Every symbol has a probability let's say $p = \frac{1}{4}$ to have an error, meaning that symbol would be replaced by a random other symbol (the distribution does not matter here as this is not formal and is purely for your understanding). For example, if the word 'play' was rolled as an error, it could become 'exfoliating' or 'mouse' or 'summer', etc.

Now given this context, Bob wants to tell Alice 'wanna play basketball' over this channel, but Alice could hear 'never play basketball'. How can we correct or detect the error? One way is by using a repetition code. This involves sending multiple copies of each symbol, and the receiving end would choose the symbol that appears most commonly in each group of 3. Now Alice hears: 'wanna never wanna play play play basketball football basketball', she decodes the message and hears 'wanna play basketball'. No more errors! However, notice this adds redundancy to the data we're sending, which makes it far less efficient. If our channel becomes noisier, let's say $p \geq \frac{1}{3}$ then our code will almost never work and we'd need to add more redundancy by sending more copies to ensure transmission is reliable enough.

Most engineers tinkered and tried to create the ideal code by varying the ratio of information

bits to redundant bits and testing the reliability, but Shannon was the only one to ask 'What is the thoeretical maximum rate of information I can have whilst still being reliable?'.

### 3.2 Exploring Reliability

The reasoning of why maximizing $R$ increases our rate of transfer is trivial, however, why does increasing the distance improve the reliability of the code? To explore this idea, let us select a random encoding and decoding function, $E$ and $D$ for every message $\mathbf{u}$ of length k. The codeword $E(u)$ is chosen randomly from the set of all codewords with uniform probabilities. The decoding function is maximum likelihood decoding, which is also known as minimum distance decoding outlined in the preliminaries section. Please follow alongside Figure 5 for the following section.

Suppose we send codeword $E(u)$ and it is received on the other end as $y$. Now draw a Hamming ball with radius $pn$ - the expected number of bit flips after transmission - and center at $E(u)$. Observe Figure 5 for a visualization of what we are doing. By Chernoff Bounds, we know that $y$ is almost certainly within our ball.

Next, draw another ball around $y$ of the same radius $pn$. The only scenario where our decoding function could fail is when there is another codeword $E(u')$ that is within a ball of radius $pn$ around $y$.

As we increase k, we also increase the number of valid codewords. This increase in density reduces the minimum distance.

Here we are viewing the same problem of trading off efficiency for reliability from a different perspective.

*Note: you can imagine y drifting away from the original codeword as it's being sent, then once received, it is matched with the closest codeword and decoded, this is the principle of what is happening above.*

## 4 Shannon's Theorem

Following from our previous discussion, we can see that this is the challenge we face in Coding Theory; how do we balance reliability with efficiency? Is it possible to create a perfectly balanced code?. In communication systems, engineers had the belief that errors during transmission were inevitable, that at most we can minimize the rate of error. Keep in mind this is not speaking in terms of the previously mentioned error probability $p$, rather, the error rate in this case is the difference between the original message and the final decoded message. The main question posed was: 'Is it possible to reliably transmit information over noisy channels, and if so, how quickly?'.

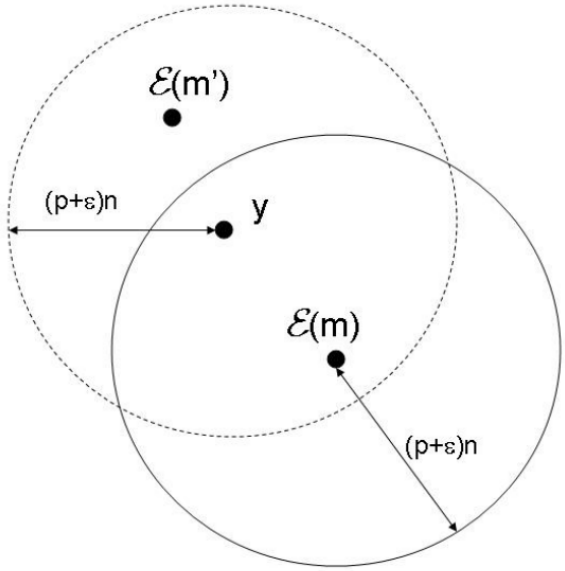



Figure 5: Hamming balls with received message $y$ and sent message $E(x)$

This is where we would like to introduce the notion of *capacity*. But before this we must build an understanding of what are *information* and *entropy*.

## 4.1 Information

In information theory, the amount of information we get from an event is defined as

$$I(X = x) = \log\left(\frac{1}{P(x)}\right) \tag{2}$$

where $P(x)$ is the probability of event $x$. This quantity can be though of intuitively as the 'surprise factor'; how surprised are you of this outcome? Trivially, if the probability of an event is almost certain, your surprise factor would be 0 and vice versa as observed in $I$. The base of the log determines the unit, either nats for base $e$ or bits for base 2. For our context, we will be using base 2 hereafter.

## 4.2 Entropy

**Entropy** of a random variable is defined as

$$H(X) = \sum_{x \in X} P(X = x) \log\left(\frac{1}{P(X = x)}\right) \tag{3}$$

Comparing this with the way we obtain the average of a random variable below:

$$\mu_x = \sum x \cdot P(x)$$

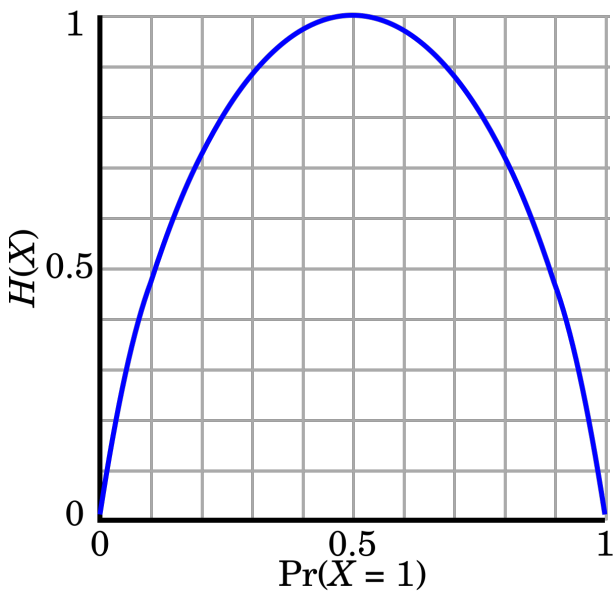



Figure 6: Entropy of a BSC

It can be intuitively understood that Entropy is the average information you get from a random variable. The more distributed the probability function, the greater the average information. It can also be thought of non-mathematically as the *randomness*.

In the case of a BSC, the plot of entropy to probability is seen in Figure 6.

For example, say you flip a fair coin once. The two events are:

$$P(H) = 0.5, \quad P(T) = 0.5$$

The entropy is:

$$H(X) = -\left[0.5 \log_2 0.5 + 0.5 \log_2 0.5\right] = -\left[0.5(-1) + 0.5(-1)\right] = 1 \text{ bit.}$$

Thus, each fair coin toss carries exactly 1 bit of information.

If the coin is biased, say

$$P(H) = 0.9, \quad P(T) = 0.1,$$

then the entropy is:

$$H(X) = -\left[0.9 \log_2 0.9 + 0.1 \log_2 0.1\right] \approx 0.47 \text{ bits.}$$

So, the biased coin has less uncertainty, hence lower entropy.

## 4.3 Mutual Information

An abstract concept following naturally from entropy is mutual information or 'how much more information do I have about random variable X if I am given Y if they depend on eachother'. Mutual information is denoted:

$$I(X;Y) = H(X) - H(X|Y) \tag{4}$$

Where $H(X|Y)$ is defined as:

$$H(X \mid Y) = -\sum_{y \in \mathcal{Y}} P(y) \sum_{x \in \mathcal{X}} P(x \mid y) \log_2 P(x \mid y). \tag{5}$$

### 4.3.1 Example with Perfect Dependence

An example will provide you with better insight and intuition on mutual information.

Consider again the fair coin toss $X$ from the previous example regarding entropy. Let

$$X = \text{Toss of a fair coin (H/T)}, \quad Y = \text{“Is the coin heads?” (Yes/No)}.$$

If you know $X$, then you know $Y$. If you know $Y$, you also know $X$.

The entropies are:

$$H(X) = 1 \text{ bit}, \quad H(Y) = 1 \text{ bit}.$$

The conditional entropy is:

$$H(Y \mid X) = 0.$$

because given X, you already know Y with certainty.

Therefore, the mutual information is:

$$I(X;Y) = H(Y) - H(Y \mid X) = 1 - 0 = 1 \text{ bit}.$$

This means if you measure X, you get 1 bit of information about Y and vice versa.

A nice visual of entropy in terms of a venn diagram is available in Figure 7.

## 4.4 Capacity

The abstract definition of capacity follows from the idea of mutual information.

**Definition 8:** Capacity is the maximum amount of information transmitted per channel use. Mathematically, it is the maximum mutual information we have on sent message x given received message y: $max \;\; I(X;Y)$

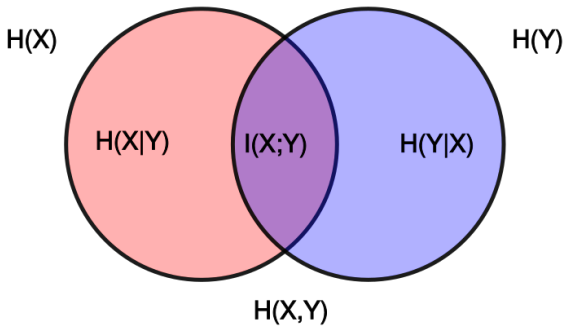



Figure 7: Venn diagram representation of entropy and information

In 1948, Claude Shannon wrote the theorem relating channel capacity with the information rate of a channel, proving that it is possible to have completely error-free communication in discrete memoryless channels so long as the information rate $R < C$ where $C$ is the 'channel capacity'.

**Theorem 1.1:** $\forall$ binary symmetric channels with $0 \leq p < \frac{1}{2}$, $\exists$ channel capacity $C$ s.t all codewords $\mathbf{x}$ can always be reliably decoded into their original message $\mathbf{u}$ with arbitrarily low error probability $Pr_e$ so long as $R < C$.

In the case of binary symmetric channels, maximum capacity is defined as: $C = 1 - H(p) = 1 + p\log_2 p + (1-p)\log_2(1-p)$ .$Pr_e$ tends to at most $2^{-\sigma n}$. where $\sigma = 1 - H(p) - R$, aka the difference between maximum and actual rate So as $n \to \infty$, the error rate approaches 0.

However, one may ask, won't the error rate approach 0 regardless of the value of R? This is a question I thought to myself and the answer is very clearly no. Assume $R > C$, then $\sigma < 0$, and so $Pr_e$ would diverge as n increases. And so the converse of the theorem is presented where, in the case $R > C$, reliable communication is impossible.
While Shannon's theorem was revolutionary, it unfortunately did not provide an empirical method to creating such efficient codes. However, by knowing this limit, engineers and mathematicians alike were able to evaluate and design their codes to be as efficient as possible while still being reliable. Examples of this include Turbo Codes (1993), LDPC codes (1990), and now Polar Codes (2008).

To feed our curiosity, let us present a very brief overview of the strategy of a geometric proof to feed the reader's curiosity and provide a general intuitive understanding. This proof is for the case of binary symmetric channels.

Take a randomly generated linear code $\mathbf{C}$ with some encoding and decoding function. Efficiency does not matter. Our decoding function will be minimum distance decoding. Let's return to our hamming space visualization in 5 and recall our explanation of error as the *drift* of the

output within its respective hamming ball. By the law of large numbers, the received codeword is almost surely going to be within $pn$ distance of the original codeword.

The number of possible sequences with exactly $pn$ flips using Stirling's approximation is

$$\binom{n}{pn} \approx 2^{nH(p)}.$$

Since the total number of binary sequences of length $n$ is $2^n$, the maximum number of disjoint Hamming balls you can pack is approximately

$$\frac{2^n}{2^{nH(p)}} = 2^{n(1-H(p))}.$$

This shows that reliable communication is possible only if the code rate satisfies

$$R < 1 - H(p).$$

# 5 Polar Codes

## 5.1 Channel Polarization - The Idea

Polar Codes are a family of linear codes that are the first family of codes to reach the Shannon Limit aka 'channel capacity'. They were created by Arikan in 2008 and later mathematically proven to reach the limit in 2013.

Arikan initially had the goal of finding a way to compress a message $u$ that is $n$ bits long. He paired up each bit in the input message as $(U, V)$. He tried encoding the bits by simply performing the XOR operation between the two bits, $(U, V) \to U \oplus V$, however, one bit of information was lost as a result. Therefore, he added another value that is linear and different from $U \oplus V$, simply choosing a copy of the second bit. So far, Arikan has encoded, $(U, V) \to (U \oplus V, V)$. Given the pair $U \oplus V$ and $V$, the operation is easily reversible, and therefore no information is lost.

One crucial property Arikan noticed is that, although he didn't succeed in compressing the information, he encoded it in a way that altered the entropy of each bit-channel (meaning each bit now has it's own probability and channel capacity). But how do the entropies of each new bit-channel change?

**Statement 1**: Given $U$ and $V$ are independant Bernoulli random variables with $0 \leq p \leq \frac{1}{2}$, we want to show that for $U \oplus V$, the new probability of error $p'$ is greater than $p$.

Given that $U$ and $V$ are mutually exclusive independant events, we can compute $p'$ like so:

$$p' = \mathbb{P}(U \oplus V = 1) = \underbrace{\mathbb{P}(U = 1, V = 0)}_{p(1-p)} + \overbrace{\mathbb{P}(U = 0, V = 1)}^{(1-p)p} = 2p(1-p).$$

Then, the difference $p' - p = \overbrace{p}^{\leq \frac{1}{2}}(1 - \underbrace{2p}_{\leq 1})$ is a positive quantity. This asserts the claim. Next,

**Statement 2**:

a) The first sub-channel's entropy increases: $H(U \oplus V) > H(U)$ and $H(U \oplus V) > H(V)$.

b) The second sub-channel's entropy decreases: $H(V \mid U \oplus V) < H(U)$ and $H(V \mid U \oplus V) < H(V)$.

From now on we denote $h(p)$ as the entropy of a Bernoulli random variable with parameter $p$. The first statement is trivial; $p' > p$ implies $h(p') > h(p)$. For the second statement, recall that the joint entropy of two independent random variables is the sum of their entropies:

$$H(U, V) = H(U) + H(V)$$

where $H(U, V)$ is the joint entropy of both $U$ and $V$. After Arıkan's transformation, we have the new pair $(U \oplus V, V)$. It is reversible to $(U, V)$, and so $H(U \oplus V, V)$ should equal $H(U, V)$. The chain rule of entropy yields

$$\underbrace{H(U \oplus V, V)}_{=2h(p)} = \overbrace{H(U \oplus V)}^{=h(p')} + H(V \mid U \oplus V)$$

[1] As a result, we can see

$$H(V|U \oplus V) = 2h(p) - h(p') < h(p)$$

As a final result, we observe how elements in the first group have entropy $H(U \oplus V) > h(p)$ and elements of the second sub-channel $H(V|U \oplus V) < h(p)$. So the first sub-channel has a larger entropy, and so its channel capacity decreases, and vice versa for the second channel. This manipulation of channel capacities is called *channel polarization* - see Figure **??**, and it's the key idea behind Polar Codes.

[1] Can you figure out why $H(U \oplus V, V) = 2h(p)$?

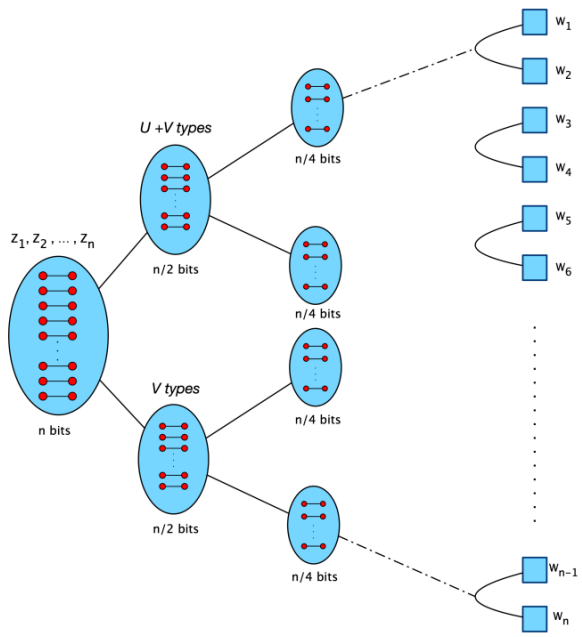
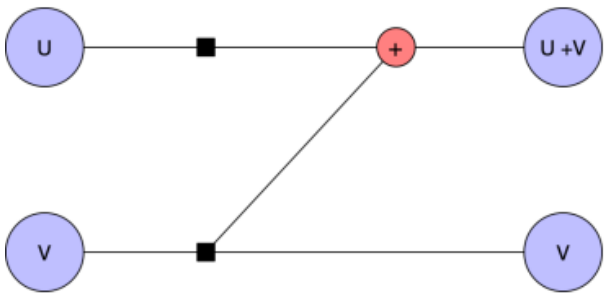



Figure 8: Polarization process




Consider what would happen if we recursively applied this algorithm. How would the channel entropies change? Let us take $2^n$ copies of the BSC, called subchannels, and recursively apply the above. We pair up $U$ and $V$, apply the operation, then separate $U \oplus V$ objects and $V$ objects whilst maintaining their order. Repeat this until only $n$ individual bits remain after separation that have either entropy 1 (useless) or entropy 0 (ideal). Observe Figure 8 for a visualization.

The final result of this stage is a set of $n$ virtual channels, either with channel capacity 0, 1 or somewhere in between. We select the channels with capacity 1 to send data through and ones with capacity 0 are *frozen*, meaning they are set to 0 and do not store meaningful data.

The careful readers may notice a glaring issue; won't there still be some 'middling' channels that have a mediocre channel capacity of 0.3, 0.6, 0.7, etc. Fortunately, Arikan proved in his paper that as number of bits $N \to \infty$, the fraction of 'middling' virtual channels vanishes, leaving channels with either channel capacity of 0 or 1.

Unfortunately, for a small number of bits, the polarization process results in some virtual channels having a mediocre capacity - 0.3, 0.6, 0.8.

### 5.2 Touching on Implementation and Advantages

As the operations are all reversible, decoding it is a simple but long algorithm. If we observe Figure 8 (the recursive polarization) we can see that $w_n$ is exclusively a V-type and does not depend on another bit channel, so it can be used to decode $w_{n-1}$, then $w_{n-1}$ can be used to decode $w_{n-2}$ and so on in a process called successive cancellation decoding which is outside the scope of this paper.

The properties of polar codes make them suitable An advantage of polar codes is that the number of steps to fully polarize the channels is a known value $\log_2 n$, since the number of copies halves at each stage.

## 6 Conclusion & Final Remarks

We've followed in the footsteps of one of the greatest revolutionaries that built the foundations of reliable and efficient communication—Claude Shannon. By exploring coding and information theory, we not only built our understanding on a number of underlying concepts but also developed a bird's eye view on how Arıkan took the teachings of Shannon to a practical application to the benefit of humanity. The ingenuity and experimentation that Arıkan expressed during his development of Polar Codes is an impressive and satisfying learning experience.

Shannon's Theorem is a beautiful mathematical construct that creates a clear guideline for engineers to build on. Polar Codes are the practical realisation of this. They are an incredible tool in the engineer's arsenal that allows optimised communication under the correct circumstances.

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# Bragg's Law in the Context of Protein Structure Determination

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**Abstract.** Scientists lacked a clear understanding of how waves interact with atoms and crystals. Analysing diffraction peaks determines the spacing between atomic planes and, consequently, the crystal's structure. The father and son team of William Henry Bragg and William Lawrence Bragg were the first to correctly analyse the diffraction peak pattern. This paper introduces Bragg's Law in the context of X-ray Crystallography. It outlines the protein structure determination by crystallisation, Fourier analysis of diffraction intensities, and finally solving the electron density map. Even in a field where mathematics is assumed to be absent, this paper illustrates the power of mathematical construction and its unforeseen applicability to complex systems and real-world phenomena.

## 1 Introduction to the fundamental problem of determining atomic-scale crystal structures

Scientists lacked a clear understanding of how waves interact with the atomic planes in a crystal. Analysing diffraction peaks determines the spacing between atomic planes and, consequently, the crystal's structure. Numerous experiments and theories have been tested before a concise explanation of the interaction between X-Ray and atoms or planes of atoms was found. The father and son team of William Henry Bragg and William Lawrence Bragg were the first to attempt determining crystal structures using X-rays. They did this in 1913, utilising X-ray diffraction to reveal how atoms are arranged within crystals. W. L. Bragg, inspired by Max von Laue and his experiments with X-Ray, proposed a corrected version of Laue's theory that he later used to prove Bragg's Law and form the basis of what we call today X-ray crystallography.

This discovery led to many of the most important scientific achievements of the last century, and these continue to the present day. This paper was the beginning of the field of X-ray crystallography, a subject that has enabled us to establish the complete structures of crystals, starting from the very simple to the most complex materials, such as proteins, viruses and the molecule that forms the very essence of life, namely DNA. Around 20 or so Nobel

Prizes have been awarded for research that has used the ideas described by Bragg. Modern genetics, medicine and the study of materials owe an incalculable debt to W. L. B., who at the astonishingly young age of 22, made a discovery that has changed our understanding of the world around us. Both father and son shared the 1915 Nobel Prize for their work, with W. L. B. remaining to the present day the youngest Nobel Prize winner ever.

## 2 Bragg's experiments and observations

His paper starts with an overview of the Laue experiment, published in [1] *Interference Phenomena with Räntgen Rays* by Herren Friedrich et al., in which a beam of X-ray was passed through a crystal of simple structure and rectilinear pencils (narrow beams or rays of X-ray with a focused and controlled nature) were observed to spread in all directions forming a geometric pattern. The spots (where the maxima points hit the photographic plate) remain symmetric to the axes of a cubical zinc blade, when the ray is perpendicular to the third axis as shown in Figure 1. After further analysis of the plate and geometric relationships between the spots and crystal structures, it was noticed that the spots are points of intersection between ellipses. Bragg explains the formation, shape and position of these ellipses with a few geometric formulas.

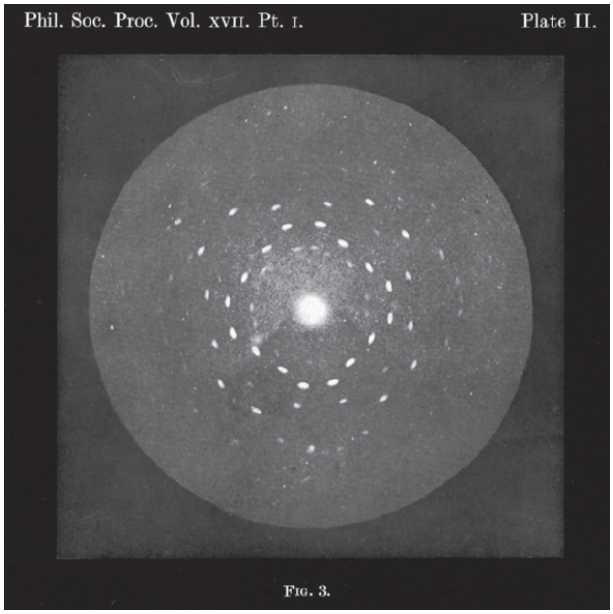


Figure 1: Plate II

Laue's theory is that secondary X-rays are emitted by molecules, and these interfere to form a pattern according to the system:

$$a\alpha = h_1\lambda$$

$$b\beta = h_2\lambda$$

$$c(1-\gamma) = h_3\lambda$$

Where $h_1, h_2, h_3$ are integers, $a$ is the side of the cube, $\lambda$ the wavelength of the secondary waves and $\alpha$, $\beta$, $\gamma$ are the directional cosines of the new maxima. Note that the waves are parallel to the $z$ axis so the surface of the beam is parallel to the $xy$ plane. Also, even though the

system is technically correct, it was made with the wrong assumption that the atoms emit secondary waves; these waves are in fact the ones reflected by the planes of the crystal, as was later proved.

This system suggests that all secondary waves are in phase in the direction whose cosines are $\alpha, \beta, \gamma$. The experimental set-up allowed the calculation of these values and proved the existence of such integers. However, not all $h_1, h_2, h_3$ are represented. Laue assumed that only certain $\lambda$ are represented because $\frac{\lambda}{a}$ also has to be an integer, but his assumption is wrong. Firstly, only certain $\lambda$ are present in the incident beam and, secondly, there are integers corresponding to certain $\frac{\lambda}{a}$ that do not appear in the pattern.

W. L. B. makes the correct assumption that a continuous range of $\lambda$ are present and then proved, with the law he formulates, why such a beam does not form a uniform darkness. He suggests that X-rays are reflected by "centres of disturbance" (i.e. planes of atoms), the same way as light, and the reflected pulse interferes with the incoming ones to build a front.

Considering planes of atoms in the crystal, the reflected beams from each plane can be seen to interfere after being reflected to a $\theta$ angle in planes found at distance $d$ of each other. The formed wavelengths are $\lambda, \frac{\lambda}{2}, \frac{\lambda}{3}, \frac{\lambda}{4}$ etc., where

$$\lambda = 2d\cos\theta$$

The intensity of a spot depends on the "average breadth" of the beam, such that if $2d\cos\theta$ is too small, pulses will neutralise each other, and if it is too big, they will be too far from each other. Thus, reflections occur for specific combinations of $\lambda, d, \theta$ and this is why the plate does not show a uniform darkness.

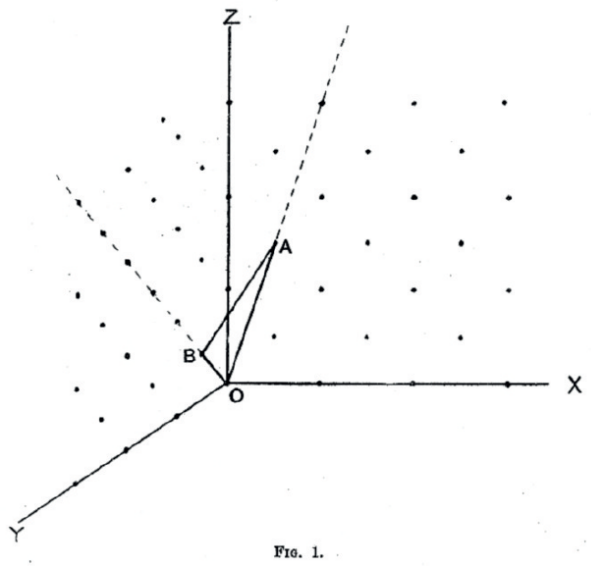


Figure 2: Crystal planes $O(0,0,0)$, $B(0,a,a)$, $A(a,0,3a)$ showing the link between the angle of incidence and the directional cosines of the plane

Designing his experiment, W. L. B. uses the Pope and Barlow theory to assume arrangements of atoms in a zinc blende and assumes that zinc and sulfur emit secondary waves of the same power to explain why some $\lambda$ are missing in Laue's model. Taking the cubic packed model, with atoms at each corner and in the centre of each face, he considers $2a$ the distance between 2

atoms and defines 2 arbitrary points in the $zx$ plane $A(pa, 0, qa)$ and in the $yz$ plane $B(0, ra, sa)$. A reflecting plane will pass through the origin and these 2 points, as shown in Figure 2. The directional cosines of such a plane are easy to calculate having defined $p, q, r, s$:

$$\vec{OC} = \vec{OA} \times \vec{OB} = (pa\ 0\ qa) \times (0\ ra\ sa) = (rq\ ps\ -pr)$$
$$l = \frac{rq}{\sqrt{p^2s^2 + q^2r^2 + p^2r^2}}$$
$$m = \frac{ps}{\sqrt{p^2s^2 + q^2r^2 + p^2r^2}}$$
$$n = \frac{-pr}{\sqrt{p^2s^2 + q^2r^2 + p^2r^2}}$$

Thus, the reflected beam will have the direction $2ln$, $2mn$, $2n^2 - 1$, calculated using the formula of the reflected beam, where $\vec{R}$ is the reflected beam and $\vec{I}$ is incident (it can be easily proven):

$$\vec{R} = \vec{I} - 2(\vec{N} \cdot \vec{I})\vec{I}$$

Notice how $n = \cos\theta$, where $\theta$ is the angle of incidence. The corresponding wavelengths satisfy $\lambda = 2d\cos\theta$, but it is easier to calculate the distance between 2 planes along the $z$ axis $l$ than the perpendicular distance $d$. Thus:

$$\lambda = 2d\cos\theta = 2l\cos\theta\cos\theta = 2ln^2$$

Another important point to be considered is the density of the atoms in the chosen plane. The strength of the reflected pulse depends on the number of atoms in a certain plane, so they need to be chosen accordingly (i.e. a restriction on 2 of the parameters $p, r, l$ defines the third). Consider the atoms as vertical columns parallel to the $z$ axis. A plane with $p = 1$ and $r = 1$ will pass through one atom in each of these rows; therefore the next such plane will be at $l = 2a$ distance from this plane. For this set of planes $\lambda = 4an^2$ and the atom density is the greatest possible. For $p = 1$ and $r = 2$, such a plane passes through atoms in half of these vertical rows, so $l$ must also be half as before to maintain the atom density constant. In general $l = \frac{2a}{\mathrm{lcm}(p,r)}$.

Furthermore, Figure 1 and Figure 3 show the pattern of spots and the planes that cause the interference. The planes that can be formed by rotation around a certain line reflect light that forms ellipses on the photographic plate. By intersecting these ellipses certain planes can be found that contain 2 specific lines.

Notice how for a reflection to be observed, the coordinates must be all odd or all even. Having $h_1, h_2, h_3$ the phase at a certain atom, W. L. B. describes

$$\frac{\lambda}{2a} = \frac{2h_2}{\sqrt{h_1^2 + h_2^2 + h_3^2}}$$

and his observations regarding this formula explain the intensity and the possible presence or absence of a certain spot, even though he admits that other explanations may also be possible. In his notes, he states that $\frac{a}{\lambda}$ must be in a certain range defined by the values of $h$ and within this range, the intensity of the spot is highest in the middle and fades towards the edges. In order to prove this, he analysed a crystal in symmetric form and when it is tilted 3°. For a

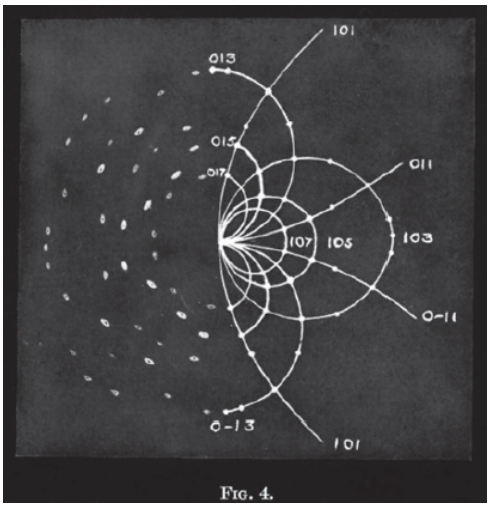


Figure 3: Intersection of dots matching ellipses corresponding to the selected planes

point in the plane defined by $(0,0,0)$, $(3a,0,a)$, $(0,3a,a)$ the corresponding spot has a higher intensity in the tilted form $\left(\frac{\lambda}{a} = 6.5\right)$ than in the symmetric form $\left(\frac{\lambda}{a} = 4.3\right)$.

It is also explained how no other arrangement of molecules offers a complete series of points and it also fits with other models for the ZnS blades.

The same analysis can be performed for other crystal arrangements to confirm these observations. He assesses a simpler crystal to show the elasticity of the X-rays and the intensity dependence on $\lambda$, not only on the position of the plane.

In Figure 4 it is shown how the spots become more elliptical as the photographic plate is moved away from the crystal, proving that light is reflected and not channelled. This is probably the observation that made W. L. B. realise the spots are formed by reflection.

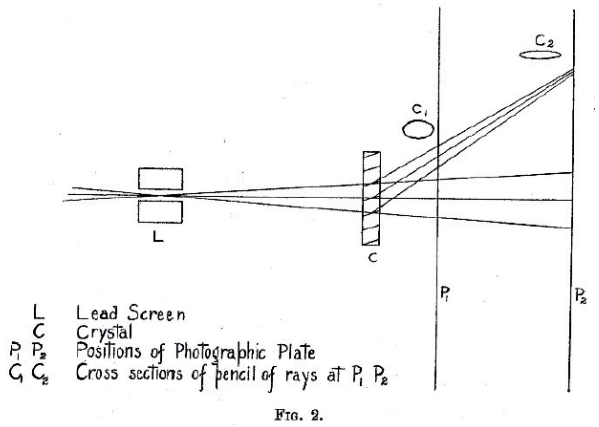


Figure 4: Experiment set-up showing the reflection of waves after passing through a crystal and the shape they acquire at increasing distance from the crystal

Further experiments with crystals show which parameters of the crystals are relevant for the formation and characteristics of the spots.

## 3 Bragg's Law: statement and proof

Bragg's Law [2] states: "when the X-ray is incident onto a crystal surface, its angle of incidence, $\Phi$, will reflect with the same angle of scattering, $\Phi$. And, when the path difference, $d$, is equal to a whole number, $n$, of wavelengths, $\lambda$, constructive interference will occur." The wavelengths that are re-emitted after hitting a plane interact with each other constructively or destructively, forming a diffraction pattern on the photographic plate. The wavelengths reflected are proportionate to the distance between the planes and depend on the angle of incidence according to the formula $n\lambda = 2d \sin \Phi$, where $d$ is the distance between planes, $\Phi$ the angle between the plane and the incident beam, and $n$ an integer (note how in his paper Bragg used the angle between the normal and the incident beam, so his original formula is $\lambda = 2d \cos \theta$).

Now, in order to prove this, multiple approaches are possible. The geometric way refers to the dependence of the phase on the wavelength of the beam and calculates the necessary angle of incidence. Another way uses Laue's equation and the condition for elastic interactions to derive Bragg's Law.

In a crystal, atoms are arranged in a certain pattern, making their interaction with the X-ray predictable. Therefore, parts of the wave are spread in a different direction, while others pass in between. For the reflected waves to create a spot on the detector, they need to interfere constructively, which happens when they are in the same phase. We know from Laue's work [3] that this happens when the path difference between different interacting atoms is an integer multiple of the wavelength. If the distance between 2 layers is $d$, the extra distance travelled is $2d \sin \Phi$. This distance must be equal to $n\lambda$. And thus, Bragg's Law.

Taking a more algebraic look at this, we need to consider $\vec{k}$ the wave vector with norm $\frac{2\pi}{\lambda}$, and $\lambda$ the wavelength of the particle. Laue's equation states:

$$\vec{k'} - \vec{k} = \vec{G}$$

where $\vec{k'}$ is the reflected wavevector, $\vec{k}$ the incident vector, and $\vec{G}$ the reciprocal lattice vector. The difference vector needs to be part of the crystal lattice for the interaction to be elastic (meaning that the two quotient spaces defined by the incident and reflected beam will be symmetric with respect to the reflective plane). Connecting this to Bragg's Law and using vector algebra:

$$\left\|\vec{G}\right\| = \left\|\vec{k'} - \vec{k}\right\|^2 = \left\|\vec{k'}\right\| + \left\|\vec{k}\right\| - 2\left\|\vec{k'}\right\|\left\|\vec{k}\right\| \cos 2\Phi$$
$$\left\|\vec{G}\right\|^2 = 2\frac{4\pi^2}{\lambda^2} - 2\frac{2\pi}{\lambda}\frac{2\pi}{\lambda} \cos 2\Phi$$
$$\left\|\vec{G}\right\|^2 = 2\left(\frac{2\pi}{\lambda}\right)^2 (1 - \cos 2\Phi) = 4\left(\frac{2\pi}{\lambda}\right)^2 \sin^2 \Phi$$
$$\left\|\vec{G}\right\| = \frac{4\pi}{\lambda} \sin \Phi$$

But, according to empirical data [4]:

$$\left\|\vec{G}\right\| = \frac{4\pi n}{d}$$

for a set of planes with distance $d$ in between. Therefore:

$$n\lambda = 2d \sin \Phi$$

Using Schrödinger's Equation offers a broader understanding of the Law and the way it is used for non-crystalline structures in modern medical research. It shows how a particle behaves after interacting with a periodic structure such as a crystal and makes a connection between the unknown structure and the known diffraction pattern. The wavelike solution of the equation shows the condition that allows diffraction to happen in the first place.

## 4 How to determine protein structure using X-ray crystallography and Bragg's Law

Using X-ray crystallography for determining structures gives mostly straightforward solutions with high precision, being a useful tool for further mechanistic understanding. Although more complex than determining a crystal structure, proteins can also be solved with the same principles. What we get from such an analysis is a diffraction pattern, as in Figure 5, and these patterns need to be interpreted. An electron density map is determined using the Fourier transformation and finally, after many refinements made using statistical methods, the protein structure is determined.

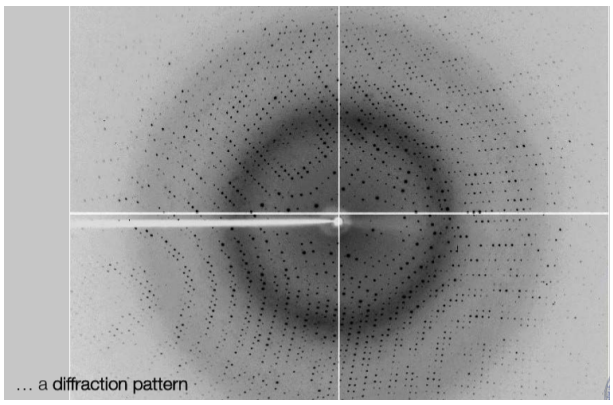


Figure 5: Diffraction pattern obtained on a photographic plate after analysing a protein

The first step is to form a protein crystal, in order to make sure the structure is coherent and the X-ray hits all the molecules from the same relative position. Next, Bragg's Law is used to determine the interplanar spacing between crystal planes. The distance is reported to the Miller indices $hkl$, which are specific to certain crystal arrangements and help solve the equation:

$$d_{hkl} = \frac{n\lambda}{2 \sin \phi}$$

The reflection angle $\phi$ is calculated by software that records the distance from the crystal to the detector $L$ and the position $x$ where the beam hits the detector. The angle can then be calculated:

$$\phi = \frac{1}{2} \arctan \frac{x}{L}$$

Now, the electron density map can be modelled via Fourier transformation. The intensity of the diffraction spot is proportional to the square of the amplitude of the scattered wave, an observation that W. L. B. also pointed out. With $F(hkl)$ being the structure factor (a complex number that represents the amplitude and the phase of the scattered X-ray), $f_j$ the atomic factor (the probability of atom $j$ to scatter X-rays), and $(x_j, y_j, z_j)$ the coordinates of atom $j$, the formula for the structure factor can be written as:

$$F(hkl) = \sum_j f_j e^{2\pi i(hx_j + ky_j + lz_j)}$$

The electron density inside the unit cell is, of course, a real-space function, and the diffraction pattern gives a Fourier transformation in its reciprocal space. This tells us how much the structure scatters the wave in each direction. The electron cloud is converted into a set of wave components that describe diffraction intensities. Performing an inverse Fourier transformation computes the real-space picture of the structure:

$$F(hkl) = \iiint \rho(x, y, z) \cdot e^{-2\pi i(hx+ky+lz)}\, dx\, dy\, dz$$

becomes

$$\rho(x, y, z) = \frac{1}{V} \sum_{hkl} F(hkl) \cdot e^{2\pi i(hx+ky+lz)+i\phi_{hkl}}$$

For the last step, the actual reconstruction of the electron density map is done by inserting the measured values of the intensities into the inverse Fourier transformation. The only problem with this method is that the phase of the beam is unknown and different models have to be tested.

Even though, theoretically, the phase problem can be resolved, it does not come from experimental data and scientists cannot be certain of its accuracy. The biggest techniques that address this problem are *molecular replacement* (which uses the model of a protein with similar structure), *multiple isomorphous replacement* (where crystals with and without heavy atoms are grown and the differences in diffraction pattern help solve the phase problem mathematically), and *anomalous dispersion* (based on the property of some atoms to scatter X-rays differently depending on the wavelength, so that data at different wavelengths form a system for solving the phases). These are approximations, as data is always lost in large analyses. In reality, the actual phase is more of a guessing game, and for this reason, statistics also play a major role in determining protein structures.

Therefore, even in a world where mathematics is assumedly absent, such as biology and biochemistry, we can see that the most fundamental concepts would have been unknown to humankind without developments in mathematics such as solutions to Schrödinger's equation, Fourier transformation, and statistics.

Bragg's Law, which seems to play a minor role in this context, is in fact the starting point of everything we know today about atomic structures, and nothing would have been possible without W. L. B. and his father's observations regarding crystallography.

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# The Qubit and Shor's Algorithm: A Step-by-Step Exploration

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**Abstract.** Shor's algorithm presents a polynomial-time quantum solution to the integer factorisation problem, threatening classical encryption protocols like RSA. This article introduces the mathematical and computational foundations required to understand the algorithm, including modular arithmetic, continued fractions, and quantum principles. We provide a step-by-step breakdown of Shor's algorithm and its quantum subroutine, emphasising accessibility and conceptual clarity, while also addressing practical constraints in current quantum hardware. This article aims to make the concepts of Shor's algorithm accessible to readers with limited background in mathematics and quantum computing, while maintaining mathematical rigour and conceptual depth throughout.

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## 0 Preliminaries.

This article has been made such that almost any person is able to understand the whole content, however, a minimum of reasoning, computer science, and mathematics concept will be required, and to ensure that the reader has these requirements, here are some definitions and concepts that we are going to use during our journey (and even some advanced definitions). The reader is not obliged to read it now, but whenever he needs to.

## 1- Number's Theory:

- **Modular arithmetic:** We say $A \equiv B \pmod{N}$, when the division of $A$ by $N$ gives a remainder of $B$. Quick example: $\frac{5}{2}$ gives a remainder of 1 $(5 = 2 \times 2 + 1)$. Thus we say: $5 \equiv 1 \pmod{2}$. If this is still a bit unclear for you, let's say that you take the number $A$ and subtract $N$ until you can't. In our previous example:

$$5 - 2 = 3, \quad 3 - 2 = 1$$

  The last number is the remainder.

- **A is a divisor of B:** $A$ is able to divide $B$, $A/B$ gives an integer.

- **The gcd of A and B (gcd(A, B)):** It's the greatest common divisor of $A$ and $B$. For instance: 24 and 18 have as common divisors: 1, 2, 3, 6. So $\gcd(24, 18) = 6$.

- **A and B are coprime:** $\gcd(A, B) = 1$. Meaning that $A$ and $B$ have no common divisors (not counting the trivial 1, which is a divisor of all whole numbers). For example, $\gcd(15, 8) = 1$, so 15 and 18 are coprime.

- **The order of A modulo N:** is the smallest number $r$ such that $A^r \equiv 1 \pmod{N}$.

- **A is semiprime:** $A$ just has two divisors (not counting 1 and $A$ itself). For example, 15 is semiprime. Another way to see it is that $A$ is the product of two prime numbers.

## 2– Computer Science:

- **Polynomial time:** We say a problem can be solved in polynomial time if: Given a problem of size $n$, the problem is solvable in a time of $n^r$, with $r$ a real number.

- **P problems:** Are the problems for which we can find solutions in polynomial time.

- **NP problems:** Problems for which we can't find solutions in polynomial time. Also explained as problems where we can easily check if a given solution is correct, but finding that solution is very hard.

- **An algorithm:** A set of instructions that is designed to accomplish a task.

- **A private key:** A key that is known only by its owner.

- **A public key:** A key that is known by everyone.

- **Converting binary to decimal:** Let's say 1101 is our binary number. You compute:

$$1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 13$$

  This helps illustrate the pattern. If not clear, a quick Google search will explain it very well.

- **Number of possible states:** Something we will use often: if we have $n$ bits, then we have $2^n$ possible states.

- **BQP problems:** BQP stands for Bounded-Error Quantum Polynomial Time. These are decision problems that can be solved by a quantum computer in polynomial time with a probability of at least $\frac{2}{3}$ of being correct. The error probability is at most $\frac{1}{3}$ for all instances.

## 3– Computer Architecture:

- **A bit:** A bit (short for binary digit) is the smallest unit of information in classical computing. It can take on one of two possible values: 0 or 1.
- **A register:** A register is a small, fast storage location inside a computer's processor that holds data temporarily. In classical computing, it typically consists of several bits and is used for arithmetic or logical operations.

## 4– Linear Algebra:

- $i$**:** Represents the elementary complex number, where $i^2 = -1$.
- **Transpose of a matrix:** Given a matrix $M$, its transpose is the matrix with rows and columns swapped. For example, here is a matrix and its transpose:

$$M = \begin{bmatrix} 1 & 2 \\ 3 & 3i \end{bmatrix}, \quad M^T = \begin{bmatrix} 1 & 3 \\ 2 & 3i \end{bmatrix}$$

- **Complex conjugate of a matrix:** Given a matrix $M$ with complex entries, its complex conjugate is basically the same matrix but negating each element that has $i$ in it:

$$M = \begin{bmatrix} 1 & 2 \\ 3 & 3i \end{bmatrix}, \quad \overline{M} = \begin{bmatrix} 1 & 2 \\ 3 & -3i \end{bmatrix}$$

## 5– Quantum Computing:

- **Physical qubits:** Physical qubits are the actual hardware qubits built using physical systems (like trapped ions or superconducting circuits).
- **Logical qubits:** Idealized qubits, error-corrected, essentially used in theory. Approximately 1 logical qubit $\approx$ 10,000 physical qubits.
- **Transformation to matrix:** Each quantum transformation can be represented as a matrix, or as its direct application on a qubit (for linear algebra adepts, you can see it as when you can represent a transformation in its matrix form or directly how it acts on the vectors).

## 6– Set's Theory:

- $\mathbb{N}$**:** Natural numbers — counting numbers like 1, 2, 3, ...
- $\mathbb{Z}$**:** Integers — all whole numbers, positive and negative, including zero: $\ldots, -2, -1, 0, 1, 2, \ldots$

- $\mathbb{Q}$: Rational numbers — numbers that can be written as a fraction $\frac{a}{b}$ where $a, b \in \mathbb{Z}$ and $b \neq 0$
- $\mathbb{R}$: Real numbers — all rational and irrational numbers (e.g., $\pi$, $\sqrt{2}$)
- $\mathbb{C}$: Complex numbers — numbers of the form $a + bi$, where $a, b \in \mathbb{R}$ and $i^2 = -1$

# 1 Introduction to Cryptography.

Cryptography is the study of mathematical techniques for securing communication and data in the presence of adversaries. Its primary goal is to ensure the confidentiality, integrity, authenticity, and non-repudiation of information. We are mainly dealing with two subjects in it: Encryption & Decryption.

**Definition 0:**

In cryptography, **encryption** (more specifically, **encoding**) is the process of transforming information in a way that, ideally, only authorized parties can decode.

## 1.1 RSA.

The RSA cryptosystem was introduced in 1977 by **Rivest**, **Shamir**, and **Adleman**, from whom it takes its name.I t is a public-key encryption scheme designed to securely encode messages over insecure communication channels by relying on the computational difficulty of factoring large composite integers. Let's discuss how it works and what is the algorithm behind it.

## 1.2 Algorithm of RSA.

Take Alice and Bob. Alice and Bob both have two big prime numbers (in general around 313 digits). Let's name the prime numbers of Alice P1 and P2, and Bob's P3 and P4.

Both Alice and Bob are going to multiply their two big prime numbers to get a new big semiprime number. Let's call them respectively SP1 and SP2.

P1 and P2 are then called the private keys of Alice, SR1 is her public key. And similarly for Bob.

If Alice wants to send something to Bob, she is going to hash her message in a way such that she uses his public key SP2, and the only way to decrypt her message, must be that her receptor knows the two big prime numbers P3 and P4, that created the public key SP2. In our case, Bob knows them, but no one else knows does. Both the public keys of Alice and Bob can be viewed by anyone, as long as her hashed message. However, due to the fact that only Bob knows what P3 and P4 are, he is the only one able to decode the hashed message.

**Note 0:**

That we are not covering how the message is hashed, and we won't. This part is not relevant for the rest of the article and thus is out of scope.

**Note 1:**

We call this an asymmetric key system because two different keys are used to send and to receive the message.

### 1.3 How to make RSA irrelevant?

The whole RSA is standing on one assumption: Finding the prime factors of a semiprime number is a NP problem. Meaning that if you have a big semiprime Number N, I can check if two prime numbers are its two prime factors (divisors). However, finding the two prime factors takes way more time, it is possible, but for example for a 313 digits semiprime number, it may take 24000 years to find them, even for a supercomputer. Thus, to be able to make RSA breakable, it is enough to show that the problem of finding the prime factors of a semiprime number is actually a P problem, so if we find a way to find efficiently factor a semiprime number, we broke the RSA system.

And this is where Shor's algorithm comes.

### 1.4 Introduction to Shor's Algorithm.

Shor's algorithm, developed by **Peter Shor in 1994**, is a quantum algorithm for efficiently factoring large integers. It runs in polynomial time on a quantum computer and poses a fundamental threat to classical cryptographic systems such as RSA, whose security relies on the hardness of integer factorization.

But before explaining how it works, you may have noticed that there is a quite scary word in its definition, "quantum". Thus, before explaining the algorithm and how it isg implemented, for the sake of the reader's comfort, I will have to give you a minimum knowledge about the world of Quantum Computing.

## 2 Introduction to the world of quantum computing.

**Note 2:**

Our goal in this article is not to focus on quantum computing, there are multiple lectures about it, and some really advanced topics will be needed during section 3.3 to explain the implementation of Shor's algorithm, and these concepts need their own article. If you feel a bit lost during the quantum computing parts, this is totally normal, but I am going to do my best to keep you on board.

**Note 3:**

While reading this article, you may encounter some intermediary algorithms, while the article will still give a short explanation of them, again our goal won't be to go deeply into them. If you would like to know more about them, a quick google search will serve you.

## 2.1 Theoritcal part.

Briefly: Quantum computers use quantum bits (qubits) instead of normal bits.

We all heard this, without really getting what is happening behind it. We also heard the famous sentence: While bits must be either in a 0 or 1 state, a qubit can be in a superposition of states $|0\rangle$ and $|1\rangle$. Some people also heard that when you measure a quantum state, you can't measure the superposition, you just randomly measure one of the possible states. At measurement, you can say that the quantum states collapse into exactly one. Let's try to discover what is really happening behind the scenes.

Formally, a quantum state (so a state "between" the $|0\rangle$ and the $|1\rangle$) is described by amplitudes $\alpha$ and $\beta$, with probability $|\alpha|^2$ of measuring 0, and probability $|\beta|^2$ of measuring 1. Thus, we have:

$$|\alpha|^2 + |\beta|^2 = 1$$

And we write for a state $X$:

$$|X\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$$

For a state with 2 qubits, instead of 1, we get $2^2$ possibilities:

$$|X\rangle = \alpha\,|00\rangle + \beta\,|01\rangle + \gamma\,|10\rangle + \delta\,|11\rangle$$

$$|\alpha|^2 + |\beta|^2 + |\gamma|^2 + |\delta|^2 = 1$$

As an example, consider the state $Y$ where:

$$|Y\rangle = \frac{1}{2}\,|00\rangle + \frac{1}{2}\,|01\rangle + \frac{1}{2}\,|10\rangle + \frac{1}{2}\,|11\rangle$$

If we try to measure the state $Y$, we have equally $\frac{1}{4}$ probability of measuring either 00, 01, 10, or 11.

## 2.2 Physical part.

But how do we create such a computer? How do we create a qubit? Usually in classical computer, we used to create a bit of 0 or 1 by using the "presence "or "absence "of electricity, but we cannot do it for quantum computers, there is no "electricity a bit present or a bit absent".

We had to find a new way to describe the 0 and the 1 state.

A widely used method to build qubits is by using the **spin of an electron**, often the electron of a phosphorus atom embedded in silicon. The two possible spin orientations — **spin-up** and **spin-down** — are used to represent the states $|1\rangle$ and $|0\rangle$ respectively. But in contrast to classical bits, the electron's spin can also be in a superposition of both up and down at once, meaning it's partly $|0\rangle$, partly $|1\rangle$. This is what allows the qubit to behave fundamentally differently from a classical bit and makes quantum computing possible.

Now that we have our foundations, we can now move the interesting part I would say.

# 3 Literally, Shor's algorithm.

Let's first state clearly the goal of the algorithm: **Given a semiprime number$N$, find the two prime numbers $P1$ and $P2$ such that: $P1 \times P2 = N$.**

The classical way to find these numbers is just by guessing randomly, also known as the brute force method. The essence or the main idea of Shor's algorithm will be that, given a random guess g, transform this bad guess into a better guess.

## 3.1 Initialization.

We start by guessing randomly a number $g$ such that $1 < g < N$. This lead to two cases:

### 3.1.1 We are lucky.

$g$ is a factor of $N$, or shares common factors with $N$. In this case, we are done thanks to Euclid's algorithm.

**Definition 1:**

**Euclid's Algorithm** is a classical method for efficiently computing the **greatest common divisor (**$GCD$**)** of two integers. It works by repeatedly replacing the larger number with its remainder when divided by the smaller, until the remainder is zero. The last non-zero remainder is the $GCD$.

**Note 4:**

Using that, to find a factor of $N$, we don't need then to guess a factor, just to find a number that shares factors with $N$.

After finding one of the prime numbers, let's assume $P1$, compute $N/P1$ and this will give you $P2$.

### 3.1.2 Most probably, we won't be lucky.

And this will make us start our process of making our bad a guess a better guess. To achieve that, we will borrow a quite helpful theorem.

**Theorem 0:**

For any two whole numbers $A$ and $B$ that are coprime, there exists a power $r$ and a multiple $m$ such that:

$$A^r = m \cdot B + 1$$

Equivalently:

$$A^r \equiv 1 \pmod{B}$$

A good example to visualize it will be $A = 2$ and $P = 5$, we find that $r = 1$:

$$2^1 \equiv 2 \pmod 5 \qquad 2^2 \equiv 4 \pmod 5 \qquad 2^3 \equiv 3 \pmod 5 \qquad 2^4 \equiv 1 \pmod 5$$

In our case, our $A$ is our bad guess $g$, and $B$ is our semiprime number $N$ that we want to factor, and ultimately, we know the existence of our $r$.

## 3.2 Making the bad guess a better guess.

Let's assume that we found r, such that $g^r \equiv 1 \pmod N$. Bring 1 to the left and you get:

$$g^r - 1 \equiv 0 \pmod N$$

And using our famous identity: $(a+b)(a-b) = a^2 - b^2$:

Replace a by g, and b by 1, and going from the right side of the equality to the left, we get:

$$\underbrace{(g^{r/2} - 1)}_{\text{P1}'} \cdot \underbrace{(g^{r/2} + 1)}_{\text{P2}'} \equiv 0 \pmod N$$

Notice that r has to be even, otherwise, r/2 won't be a whole number. If we found an r odd, we repeat our algorithm from the beginning.

What we found now is actually really close to what we want, instead of finding the two numbers P1 and P2 such that $P1 \times P2 = N$, we found that N divides the product of two other numbers P1′ and P2′. Which can be written as:

$$P1' \times P2' = mN \quad , m \in \mathbb{N}$$

Even more precisely, we found two numbers P1′ and P2′ that share factors with N. Thus:

$$\gcd(\text{P1}', N) = \text{P1 or P2. (We can replace P1}' \text{ by P2}', \text{ the outcome stays the same).}$$

And remember that thanks to our beautiful Euclide's algorithm, we are able to compute

$$\gcd(\text{P1}', N) = P1$$

easily. Compute then $\frac{N}{\text{P1}} = \text{P2}$, (and vice versa if we got that $\gcd(\text{P1}', \text{N}) = \text{P2}$). And we have successfully found our P1 and P2.

*End of our algorithm.*

Or maybe not?

You may feel that this is too easy, in the sense where, using this algorithm we could directly break the RSA, why are we using it? Remember also that I gave you an introduction about quantum computing, but I didn't talk at all about it at any point.

Let's recall the algorithm's main steps.

1 – Make a guess, $g < N$ that shares no factors with $N$.

2 – Find $r$ such that $g^r = mN + 1$.

3 – If $r$ is even, calculate $(g^{r/2} + 1)$ and $(g^{r/2} - 1)$. If $r$ is odd, go back to step 1.

4 – Use Euclid's algorithm to find the greatest common divisor.

Where is the **quantum** here?

## 3.3 The quantum subroutine of Shor's algorithm.

Notice that we didn't discuss before how we are finding r, which is in fact that hardest (probably due to the quantum concepts) part of the whole algorithm.

**Note 5:**

Through this part, the quantum part of the explanation (which is still the main thing) is going to be a bit simplified, essentially when we come to some deep notions such as Quantum Fourier Transform, etc. Such quantum concepts as said before need even their own whole article. And as again mentioned before, one of the primary goals of the article is to keep the reader on board. There will be some analogies, to the classical architecture of a computer, for the sake of your understanding, please go back to section 0 and give a quick overview to the preliminaries again.

### 3.3.1 Initialization.

To achieve this quantum subroutine, we will need two quantum registers, as normal registers that we have in normal computers, these ones are just going to contain qubits instead of normal bits.

First register: is going to be 2n qubits, initialized as:

$$|0\rangle^{\otimes 2n}$$

The number $n$ just determines how accurate we want our calculations to be, however, the 2nd register size depends on the first one.

Second register is going to be n qubits, initialized to: $|1\rangle$

Note that $\otimes$ in context of quantum computing doesn't mean the XOR gate as a lot of people would think. It is a tensor product. Just consider it as a normal multiplication but between the quantum states.

Note also that for the first register, we mean that all the bits are set to 0, meaning that the state of that register is: $|000\ldots000\rangle$.

But for the second register, we mean that the value of that register is one, meaning that the state of that register is: $|000\ldots001\rangle$. It is just a story of notations.

What does it do? Applying the Hadamard Transform to a $|1\rangle$ or $|0\rangle$ gives this:

**Example 1.** Applying this to $0\rangle$ and $1\rangle$ gives

$$H\,|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$
$$H\,|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

### 3.3.2 Hadamard Transform.

Let's first define how the Hadamard Transform works. The Hadamard gate is formally defined as this:

$$H = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

What does it do? Applying the Hadamard Transform to a $|1\rangle$ or $|0\rangle$ gives this:

$$H\,|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$
$$H\,|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

During this part, we will be interested essentially in its application on the 0 state, since our first register is just full of 0's.

Notice that on 0, it created exactly a state such that the probability of getting one of the states $x$ such that $x$ in $\{0, (2^1 - 1)\}$ (converting the decimal to binary) is exactly equal.

Let's try to apply it to 0 states with 2 qubits.

$$H^{\otimes 2}\,|00\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$$

And on 3 qubits?

$$H^{\otimes 3}\,|000\rangle = \frac{1}{\sqrt{8}}(|000\rangle + |001\rangle + |010\rangle + |011\rangle + |100\rangle + |101\rangle + |110\rangle + |111\rangle)$$

Did you notice the pattern? If not let me show it to you:

Whenever we apply the H gate to a register of size $n$, such that all its qubits are 0, we get a state such that the probability of measuring a state in $\{0, 1, 2, \ldots, 2^n - 1\}$ is exactly equal for all of them.

You may wonder now why we are doing that, we will see why soon.

To end this part, we will formally say that the state of our first register now is:

$$\frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle$$

### 3.3.3 $U$.

We will apply another transformation, but to the 2nd register. Let's call this transformation $U$. $U$'s definition is part of the algorithm, meaning that it is not a famous transformation created from before. Formally, the definition of $U$ is:

$$U|k\rangle = |g \cdot k \bmod N\rangle$$

With $g$ the random guess that we picked before.

Even more formally:

$$U|k\rangle = \begin{cases} |k\rangle & \text{if } 0 \leq k < N \\ |g \cdot k \bmod N\rangle & \text{if } N \leq k < 2^n \end{cases}$$

One good property of $U$ is that $U$ is a unitary transformation: which means that if you take the transpose then the complex conjugate of its matrix representation, it gives you $U$ back.

How are we going to use $U$? First let's look at our state now.

The quantum state of our full system (including the 1st and the 2nd register) is the tensor product $\otimes$ of both first and second register. As I said $\otimes$ is just a normal multiplication of the states. Our first register is now a sum of equal states, and our second register is still just in a 1 state. "Expanding" the expression:

$$\left(\frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle\right) \otimes |1\rangle$$

gives:

$$\frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} (|x\rangle \otimes |1\rangle)$$

Don't overthink about it, it is the exact same way as you would expand $(a+b+c)d = ad + bd + cd$. For each x (the values in each state from 0 to $2^n - 1$), we apply the transformation x times on the qubit at state 1 associated to it. It is a bit hard to understand it like that, we can use a visualization to understand it more:

For each pair $|x\rangle \otimes |1\rangle$, we apply U getting:

$$|x\rangle \otimes |1\rangle \longrightarrow |x\rangle \otimes U^x |1\rangle = |x\rangle \otimes |g^x \bmod N\rangle$$

As an example, applying U to a state $|101\rangle \otimes |1\rangle$ (101 read as $4 + 0 + 1 = 5$, recall the preliminaries) gives:

$$|101\rangle \otimes U^5 |1\rangle = |101\rangle \otimes |g^5 \bmod N\rangle$$

And again, g is our random guess that we picked before.

Notice that when applying the transformation, we will get a certain periodicity in our quantum system due to modulo properties. To visualize it even better, let us consider that our x is

going from 0 to 7 (using thus 3 qubits), that our N was 26, and our random guess was 3, after applying all what we had before, we will get:

$$\frac{1}{\sqrt{8}}(|000\rangle\,|3^0 \bmod 26\rangle + |001\rangle\,\left|3^1 \bmod 26\right\rangle + |010\rangle\,|3^2 \bmod 26\rangle +$$

$$|011\rangle\,|3^3 \bmod 26\rangle + |100\rangle\,\left|3^4 \bmod 26\right\rangle + |101\rangle\,|3^5 \bmod 26\rangle +$$

$$|110\rangle\,|3^6 \bmod 26\rangle + |111\rangle\,|3^7 \bmod 26\rangle)$$

Which exactly gives:

$$\frac{1}{\sqrt{8}}(|000\rangle\,|1 \bmod 26\rangle + |001\rangle\,|3 \bmod 26\rangle + |010\rangle\,|9 \bmod 26\rangle +$$

$$|011\rangle\,|1 \bmod 26\rangle + |100\rangle\,|3 \bmod 26\rangle + |101\rangle\,|9 \bmod 26\rangle +$$

$$|110\rangle\,|1 \bmod 26\rangle + |111\rangle\,|3 \bmod 26\rangle)$$

**Note 6:**

$|x\rangle \otimes |y\rangle$ is the same as $|x\rangle\,|y\rangle$

Notice also that, the period of the periodicity is 3, which is exactly the number $r$ that we are searching for, $3^3 \equiv 1 \pmod{26}$.

Thus, if we are able to extract the periodicity from the state, we would be done.

### 3.3.4 The periodicity, aka r.

This is actually a way harder task than expected. When you see the representation of the state in front of you it seems obvious, however, as we said before, if we just measure now our state it is just going to give us a random state since they all have equal probability. Which is not what we want. In fact, we are not able to extract the periodicity. However, we have something called the QFT (for **Q**uantum **F**ourier **T**ransform).

The QFT is formally defined as:

$$\text{QFT}\,|x\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{\frac{2\pi i x k}{N}}\,|k\rangle$$

Where N is the number of qubits that represent x.

The QFT doesn't extract periodicity, however, it is able to extract the frequency. And remember that

$$\text{Period} = \left(\frac{1}{\text{frequency}}\right)$$

Without entering too much into the details, believe me, the QFT would need a whole article for itself, briefly: The QFT in Shor's algorithm transforms the hidden period r of $g^x \bmod N$ into measurable frequency information, allowing us to extract r from a single quantum

measurement.

Basically, the amplitudes (that represent the probability of being measured) C are **peaked** near j/r, where j is just the state that we randomly measured, since a frequency function just works with adding $1/r$ each time.

Thus, since the highest probability is near to j/r, we most probably are going to measure j/r.

But now yes we have j/r, how do we get r? If you measure $j/r = 0.312$, how can we extract r?

Luckily we already have a classical (runnable on a classical computer) algorithm that can extract r.

### 3.3.5 Continued Fraction Algorithm (CFA).

The best way to explain the Continued Fractions Algorithm is to give an example.

Assume that we found that j/r = 0.312

We start by writing it as a fraction:

$$0 + \frac{312}{1000}$$

We then reverse the fraction and get this:

$$= 0 + \frac{1}{\frac{1000}{312}} = 0 + \frac{1}{3 + \frac{8}{39}}$$

And we repeat the same process until we get a 1 in the numerator.

$$0.312 = 0 + \frac{1}{3 + \frac{1}{4 + \frac{1}{1 + \frac{1}{7}}}}$$

For each fraction, we get a better approximation for 0.312:

$$0.312 \approx \frac{1}{3}, \quad j = 1, \quad r = 3$$

$$0.312 \approx \frac{1}{3 + \frac{1}{4}} = \frac{4}{13}, \quad j = 4, \quad r = 13$$

$$0.312 \approx \frac{1}{3 + \frac{1}{4 + \frac{1}{1}}} = \frac{5}{16}, \quad j = 5, \quad r = 16$$

We choose an approximation of r such that it follows two conditions:

1- It should trivially be less than N

2- It should be an even number.

If none of our approximations gives a valid r, we repeat the same process from the beginning.

And this gives an end to our quantum subroutine.

And an official end of our Shor's algorithm.

## 4 Final thoughts: does Shor's algorithm really break RSA?

During your lecture, you went through all the Shor's algorithm, you had a whole journey with it, and I hope that I was able to make you enjoy the journey. You probably have a lot of questions in your mind now, and one of them would be that you saw with your own eyes that we have an algorithm that breaks our usual encryption. So how is our data safe?

Actually, we not really did. We are assuming that we have a quantum computer that is able to run Shor's algorithm, and we don't. Shor's algorithm would need 4100 logical qubits, or 20 000 000 physical qubits to run.

Currently, the main actor in this field is IBM, and they are aiming for 1121 physical qubits soon, far from our 20 000 000 ones. But the day we will be able to make a quantum computer with 20 000 000 physical qubits, we will definitely be able to break RSA, so the question is not if we are going to be able to, but **when**?

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